

Worksheet 16

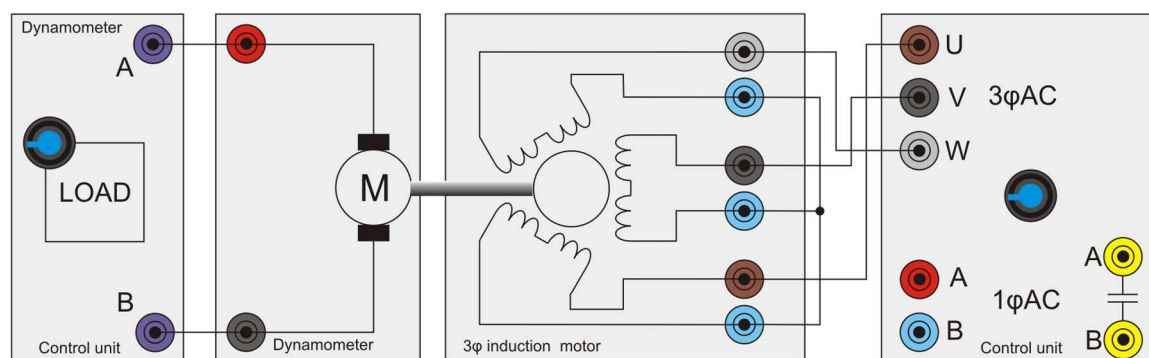
Three-phase motor characteristics

Modern electrical machines



This photograph shows a section of a three phase motor with the outer shell cut away. Note the lack of brushes and the lack of a winding in the rotor.

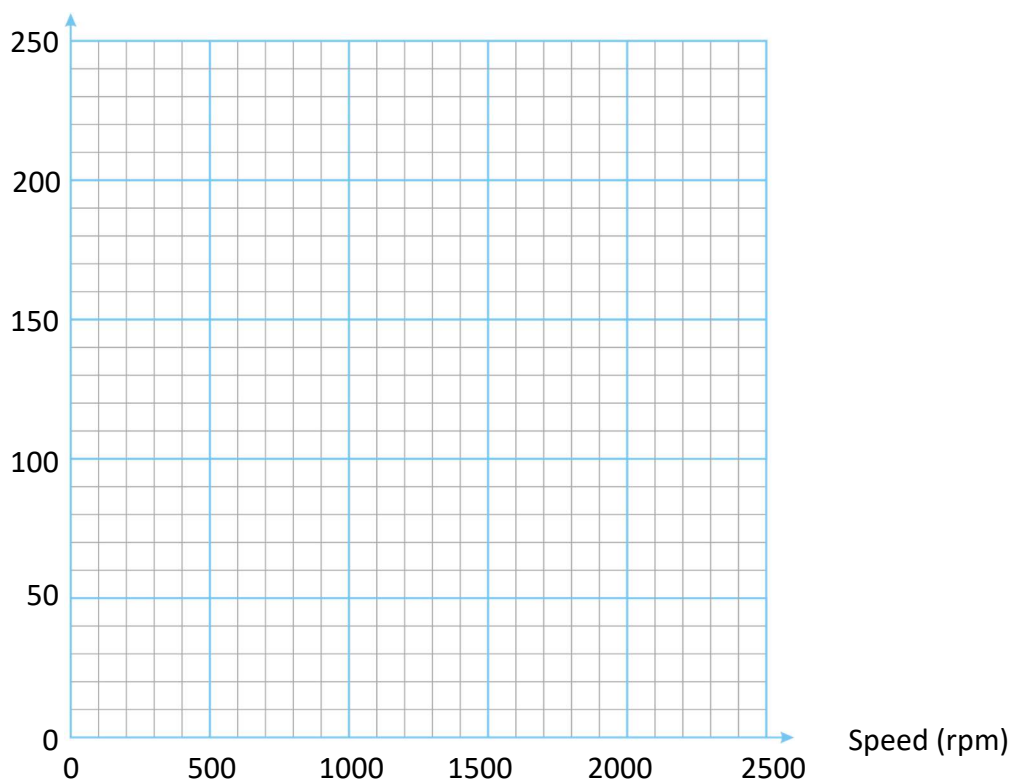
Induction motors work by inducing a current, and magnetic field, into the rotor which has a slightly different phase to the driving current/field.



Over to you:

- 1) Set up the three phase induction motor as shown in the diagram above.
- 2) On your PC, load 'Open Log_3phase.bat'.
- 3) Use settings SINE, 20V, IMPROVED.
- 4) With an output frequency of 50Hz, run the program and take an automatic set of readings over a range of dynamometer loads.
- 4) Repeat this for output frequencies of 60, 70, 80, 90, 100 Hz.
- 5) If you can plot all output lines on the same graph.
- 6) On the next page, sketch the results .

How do you change direction of rotation?



How does it compare for efficiency? Research the reason for this and give your findings below.

[illegible]